

SUPPORT FAMILIES OF CLOSED SUBSPACES OF ℓ^2 , WITH AN APPLICATION TO DIAGONALIZABLE PROJECTION FILTERS

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ABSTRACT. For a closed subspace M of $\ell^2(\mathbb{N})$ let $\mathcal{F}(M)$ be the family of supports of its vectors. We determine which families arise this way in finite rank and record what survives in infinite rank. Closedness forces $\mathcal{F}(M)$ to be closed under arbitrary unions, and Gaussian elimination among minimal supports is the circuit elimination axiom; combined with a theorem of Aroca, Bossinger, Falkensteiner, Garay López, González-Ramírez and Valencia Negrete, this gives, over $\mathbb{C}$, exactly the scrawl families of cofinitary matroids on $\mathbb{N}$ with $\mathbb{C}$ -representable duals of finite rank. Of the three Bowler–Carmesin scrawl axioms, union closure holds outright and elimination holds for finitely many deleted coordinates; a model space K_Θ shows the maximality axiom can fail, and can fail so far that $\mathcal{F}(K_\Theta)$ is the scrawl family of no matroid. The support family also computes the interaction of M with the diagonal masa: the trace of a maximal filter of the projection lattice on the diagonal block is $\bigcap_M \mathcal{D}(M)$, matroid independence is lattice-theoretic disjointness from the block, and M belongs to a maximal filter with ultrafilter trace exactly when the covering number $\chi(M)$ of its support matroid is infinite. This invariant grades a property studied in the companion papers: a subspace is intimate exactly when $\chi \geq 3$. The grading is strict — Gowers’s intimate subspace has $\chi = 3$, so it lies in no diagonalizable maximal filter — but collapses on filters, every member of a diagonalizable maximal filter having $\chi = \infty$.

1. INTRODUCTION

For $x \in \ell^2(\mathbb{N})$ write $\text{supp}(x) = \{n : x_n \neq 0\}$, and for a closed subspace $M \subseteq \ell^2(\mathbb{N})$ put

$$\mathcal{F}(M) = \{\text{supp}(x) : x \in M\} \subseteq 2^{\mathbb{N}}.$$

Question 1.1. Which families $\mathcal{F} \subseteq 2^{\mathbb{N}}$ arise as $\mathcal{F}(M)$?

The answer in finite rank is a piece of matroid representability (Section 5). In infinite rank it is open, and Section 6 locates the obstruction in a question about coefficient vanishing in model spaces.

The reason for asking is lattice-theoretic, and it is the reason this paper belongs with [FWI, FWII, FWIII, FWIV]. Write $\mathcal{P}(\ell^2)$ for the lattice of closed subspaces, and let $\{\ell^2(S) : S \subseteq \mathbb{N}\}$ be the projection lattice of the diagonal masa — a copy of $2^{\mathbb{N}}$, and a maximal Boolean subalgebra, a *block* in the language of quantum logic [K83]. A *pluf* is a maximal filter in $\mathcal{P}(\ell^2)$ [FWI]; its trace on the block is $\mathcal{S}(\pi) = \{S : \ell^2(S) \in \pi\}$, and the question of when that trace is an ultrafilter is the diagonalization question of [FWII]. Section 7 shows that the trace is computed from support families and from nothing else, that matroid independence is disjointness from the block, and that a single invariant — the covering number of the support matroid — decides whether M can sit in a pluf with ultrafilter trace.

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Section 8 draws the consequences for the companion papers. The covering number grades the intimacy of [FWII]: a subspace is intimate exactly when $\mathbb{N}$ is not the union of two sets S with $M \wedge \ell^2(S) = 0$, that is, exactly when $\chi(M) \geq 3$, while membership in a diagonalizable pluf requires $\chi(M) = \infty$. The two conditions differ on subspaces: Gowers's intimate subspace of [FWII] has $\chi = 3$ (Theorem 8.4), so it belongs to no diagonalizable pluf. They agree on plufs: every member of a pluf with ultrafilter trace has $\chi = \infty$ (Proposition 8.2), so "every member intimate" and "every member diagonally consistent" define the same class of filters. Section 8 closes with a constraint on the rigidity conjecture of [FWII]: every addable blocker has infinite covering number.

Conventions and the choice of field. $\ell^2 = \ell^2(\mathbb{N})$ is taken over $\mathbb{C}$. Two of the results below are field-sensitive and one is not; we fix which is which. Union closure (Section 2) needs only that there are more scalars than coordinates, and holds over $\mathbb{R}$ and $\mathbb{C}$ alike; over a finite field it fails, and Section 5 shows the failure is not an artifact of the argument. The finite-rank classification is representability over the field in question, and representability genuinely depends on it (Question 9.6); the theorem of [ABFGGV] that we invoke carries a cardinality hypothesis on the field which $\mathbb{C}$ satisfies. Everything in Sections 7–8 — the Galois connection, the trace formula, the covering criterion, and the consequences for [FWII] — uses no property of the scalars at all and holds verbatim over $\mathbb{R}$. The companion papers work over $\mathbb{R}$, where their formal verification is carried out; the results transferred here are among those to which the field is immaterial.

Members of $\mathcal{F}(M)$ include $\emptyset = \text{supp}(0)$. For $S \subseteq \mathbb{N}$ we write $\ell^2(S) = \{x : \text{supp}(x) \subseteq S\}$, which is $\mathcal{H}_S$ in the notation of [FWII].

2. CLOSURE UNDER UNIONS

Lemma 2.1. *Let M be any linear subspace of ℓ^2 and $x, y \in M$. Then $\text{supp}(x) \cup \text{supp}(y) \in \mathcal{F}(M)$.*

Proof. For $\lambda \neq 0$ the coordinate $(x + \lambda y)_n$ vanishes for at most one λ , namely $-x_n/y_n$ when $y_n \neq 0$. There are countably many coordinates and uncountably many scalars, so some λ avoids every bad value, and then $\text{supp}(x + \lambda y) = \text{supp}(x) \cup \text{supp}(y)$. $\square$

The proof uses that there are more scalars than coordinates. The infinite version uses closedness, and uses it once.

Proposition 2.2 (Countable unions). *Let $M \subseteq \ell^2$ be closed and $x^{(1)}, x^{(2)}, \dots \in M$. Then there is $x \in M$ with $\text{supp}(x) = \bigcup_i \text{supp}(x^{(i)})$.*

Proof. Normalize $\|x^{(i)}\|_2 = 1$, so $\|x^{(i)}\|_\infty \leq 1$. We choose $c_i \neq 0$ recursively and set $x = \sum_i c_i x^{(i)}$, writing $s_m = \sum_{i \leq m} c_i x^{(i)}$.

At stage m , two constraints bind. First, to keep $\text{supp}(s_m) = \text{supp}(s_{m-1}) \cup \text{supp}(x^{(m)})$, the coefficient c_m must avoid $-(s_{m-1})_n/x_n^{(m)}$ for each of the countably many n with $x_n^{(m)} \neq 0$. Second, having chosen c_m , put

$$\delta_m = \min\{|(s_m)_n| : n \leq m, (s_m)_n \neq 0\} > 0,$$

a minimum over a finite set, and require of all later coefficients that $|c_i| \leq 2^{-(i-m)-1}\delta_m$ for $i > m$, together with $|c_i| \leq 2^{-i}$ throughout. The admissible set for c_m is a punctured disc less a countable set, so the recursion runs.

Since $\sum_i |c_i| \|x^{(i)}\|_2 < \infty$ the series converges in ℓ^2 ; the partial sums lie in M and M is closed, so $x \in M$. This is the only step that uses closedness. Every term is supported

in $\bigcup_i \text{supp}(x^{(i)})$, so $\text{supp}(x)$ is contained in that union. Conversely, if $n \in \text{supp}(x^{(m)})$ then $n \in \text{supp}(s_m)$, so $|(s_m)_n| \geq \delta_{\max(m,n)}$ once the index is large enough for n to be frozen, and the tail $\sum_{i>\max(m,n)} |c_i|$ is at most half of it; hence $x_n \neq 0$. $\square$

Corollary 2.3. $\mathcal{F}(M)$ is closed under arbitrary unions, and $\sigma(M) := \bigcup \mathcal{F}(M)$ is itself a member of $\mathcal{F}(M)$.

Proof. An arbitrary union of supports is a union of countably many of them: a subset of $\mathbb{N}$ that is a union of members of $\mathcal{F}(M)$ is the union of countably many, one through each of its points. Apply Proposition 2.2. $\square$

Closedness is not decoration here. The subspace of finitely supported sequences orthogonal to a fixed vector of full support is a non-closed subspace whose support family omits its own union.

3. ELIMINATION

Nonemptiness and union closure do not suffice.

Example 3.1. $\mathcal{F} = \{\emptyset, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$ contains $\emptyset$ and is closed under unions, but is not $\mathcal{F}(M)$: vectors x, y with supports $\{1, 2\}$ and $\{1, 3\}$ have $x_1, y_1 \neq 0$, and $z = x - (x_1/y_1)y$ has support $\{2, 3\}$.

Proposition 3.2 (Elimination). *Let $A \neq B$ be nonempty members of $\mathcal{F}(M)$ and $p \in A \cap B$. Then $(A \cup B) \setminus \{p\}$ contains a nonempty member of $\mathcal{F}(M)$. Minimality of A and B is not required; the case of minimal A, B is the circuit elimination axiom.*

Proof. Eliminate p as in Example 3.1. The result is nonzero, since $z = 0$ would make x a multiple of y and force $A = B$, and is supported in $(A \cup B) \setminus \{p\}$. $\square$

Nonemptiness, union closure and elimination among minimal members are the axioms of a structure introduced for unrelated reasons, to which we now turn.

4. CIRCUITS AND SCRAWLS

We use only the following. A *matroid* on a ground set E is given by its family of *circuits*: a nonempty antichain $\mathcal{C} \subseteq 2^E$ satisfying circuit elimination — for distinct $C, C' \in \mathcal{C}$ and $p \in C \cap C'$, some circuit lies in $(C \cup C') \setminus \{p\}$ — together with the requirement that the *independent* sets, those containing no circuit, satisfy the maximality axiom (SM): every independent subset of a set X extends to a maximal independent subset of X . For finite E the last condition is automatic; for infinite E it is the axiom isolated by Bruhn, Diestel, Kriesell, Pendavingh and Wollan [BDKPW13]. A *scrawl* is a union of circuits, and the scrawl family is closed under arbitrary unions. Bowler and Carmesin [BC18] axiomatize matroids by their scrawls: (S1) union closure; (S2) scrawl elimination, the multi-coordinate form of circuit elimination; and (SM).

For a matrix A over a field K with columns indexed by E , the supports of the vectors of $\ker A$ are the scrawls of the matroid whose circuits are the minimal dependencies among the columns. Question 1.1 asks which scrawl families are realized when $K = \mathbb{C}$, $E = \mathbb{N}$, and the kernel is required to be norm-closed in ℓ^2 .

Theorem 4.1. *Let $M \subseteq \ell^2(\mathbb{N})$ be closed. Then $\mathcal{F}(M)$ satisfies (S1), and satisfies (S2) for finite deleted sets: if $w = \text{supp}(x)$ with $x \in M$, if $X \subseteq w$ is finite, and if $u_p \in M$ for $p \in X$*

have supports w_p with $p \in w_q \iff p = q$, then for every $z \in w \setminus \bigcup_{p \in X} w_p$ there is $y \in M$ with

$$z \in \text{supp}(y) \subseteq \left(w \cup \bigcup_{p \in X} w_p \right) \setminus X.$$

Proof. (S1) is Corollary 2.3. For the elimination, $a_p = x_p/(u_p)_p$ is well defined by the hypothesis on the w_q , and $y = x - \sum_{p \in X} a_p u_p$ — a finite sum, hence in M — has $y_p = 0$ for $p \in X$. No subtracted term touches z , so $y_z = x_z \neq 0$. $\square$

For infinite X the correction term is an infinite series that must converge in ℓ^2 , and there is no freedom to arrange it: rescaling u_p rescales a_p inversely, so $a_p u_p$ is determined by the data. Question 9.2 records the gap. Axiom (SM) is the subject of Section 6.

5. THE FINITE-RANK ANSWER

Theorem 5.1 (Aroca et al. [ABFGGV, Thms. 2.15, 3.5]). *Let K be a field, E a set, and $\{0\} \neq W \subseteq K^E$ finite dimensional. Write $\mathfrak{T}(W)$ for the supports of elements of W . Then the minimal nonempty members of $\mathfrak{T}(W)$ are the circuits of a matroid on E and every member of $\mathfrak{T}(W)$ is a union of circuits; and if $\#E < \#K$ then $\mathfrak{T}(W)$ is exactly the scrawl family of that matroid.*

The cardinality hypothesis is sharp. Over $\mathbb{F}_2$, the span of $(0, 1, 1)$ and $(1, 0, 1)$ in $\mathbb{F}_2^3$ has support family $\{\emptyset, \{2, 3\}, \{1, 3\}, \{1, 2\}\}$, which is not union-closed: there are too few scalars to run Lemma 2.1. For $E = \mathbb{N}$ and $K = \mathbb{C}$ the hypothesis reads $\aleph_0 < 2^{\aleph_0}$.

Theorem 5.2. *Let $M \subseteq \ell^2(\mathbb{N})$ be a nonzero closed subspace of finite codimension in some $\ell^2(S)$, or more generally a closed subspace whose orthogonal complement within its own support block is finite dimensional. Then $\mathcal{F}(M)$ is the scrawl family of a matroid on $\mathbb{N}$ whose dual is $\mathbb{C}$ -representable of finite rank; conversely every such scrawl family arises. In particular every finite restriction of every support family of a closed subspace is the scrawl family of a representable matroid.*

Proof. A closed subspace of finite codimension in $\ell^2(S)$ is the kernel of a finite matrix over $\mathbb{C}$ acting on $\ell^2(S)$, so Theorem 5.1 applies with $E = S$, $K = \mathbb{C}$; extend by $\emptyset$ off S . Conversely a representable dual of finite rank presents M as such a kernel, and a kernel of finitely many continuous functionals is closed. For the last sentence, restrict to a finite set of coordinates: the restriction of $\mathcal{F}(M)$ is the support family of the image of M under the coordinate projection, a finite-dimensional space over $\mathbb{C}$. $\square$

By Mayhew, Newman and Whittle [MNW14], following Vámos [V78], the class of representable matroids is not finitely axiomatizable, so the characterization in Theorem 5.2 is decidable [S87] but admits no finite list of axioms.

6. INFINITE RANK: A MODEL SPACE

Lemma 6.1. *Let $\mu_1 < \dots < \mu_m$ be real and let $a \in \mathbb{R}^m$ be nonzero. Then $F(t) = \sum_{k \leq m} a_k e^{\mu_k t}$ has at most $m - 1$ real zeros.*

Proof. Induct on m , discarding vanishing coefficients so that all $a_k \neq 0$. For $m = 1$, F has no zeros. For $m > 1$, suppose F had m zeros. Then $G(t) = e^{-\mu_1 t} F(t)$ has the same m zeros, so by Rolle's theorem G' has at least $m - 1$; but $G'(t) = \sum_{k \geq 2} a_k (\mu_k - \mu_1) e^{(\mu_k - \mu_1)t}$ is an exponential sum with $m - 1$ nonzero coefficients and distinct exponents, hence has at most $m - 2$ zeros. $\square$

Example 6.2. Choose distinct $\lambda_1, \lambda_2, \dots \in (0, 1)$ with $\sum_k (1 - \lambda_k) < \infty$ — the summability is used only for the identification below, not for the support claim — put $v^{(k)} = (\lambda_k^n)_{n \geq 0}$ and $M = \overline{\text{span}}\{v^{(k)}\}$. Then every cofinite subset of $\mathbb{N}_0$ lies in $\mathcal{F}(M)$.

Proof. Fix finite $S \subseteq \mathbb{N}_0$ and put $m = |S| + 1$. Applying Lemma 6.1 with $\mu_k = \log \lambda_k$ shows that a nontrivial combination of $v^{(1)}, \dots, v^{(m)}$ vanishes at no more than $m - 1$ points of $\mathbb{N}_0$; in particular the $v^{(k)}$ are linearly independent, so their span V_m has dimension m . Evaluation at the $|S| = m - 1$ points of S is a linear map $V_m \rightarrow \mathbb{R}^{m-1}$, so some nonzero $x \in V_m$ vanishes on S . That x has at least $m - 1$ zeros on $\mathbb{N}_0$ and at most $m - 1$, so its zero set is exactly S and $\text{supp}(x) = \mathbb{N}_0 \setminus S$. $\square$

Under $\ell^2(\mathbb{N}_0) \cong H^2(\mathbb{D})$ the vector $v^{(k)}$ is the reproducing kernel at λ_k and M is the model space $K_\Theta = H^2 \ominus \Theta H^2$ for the Blaschke product Θ with zeros λ_k ; members of $\mathcal{F}(M)$ are Taylor-coefficient supports of functions in K_Θ [GMR16].

What Example 6.2 establishes is that no cofinite set is a minimal nonempty member of $\mathcal{F}(M)$, since every cofinite set properly contains another. Whether $\mathcal{F}(M)$ contains any set of infinite complement, and hence whether it has any minimal nonempty member at all, is Question 9.3. We avoid the word *circuit* for the minimal members here: that terminology presupposes that $\mathcal{F}(M)$ is a scrawl family, which is what is in doubt.

Suppose K_Θ contains no nonzero function with infinitely many vanishing Taylor coefficients. Then every nonzero support is cofinite and $\mathcal{F}(M) = \{\emptyset\} \cup \{\text{cofinite sets}\}$, and $\mathcal{F}(M)$ is the scrawl family of no matroid — for a reason weaker than the failure of (SM). Were $\mathcal{F}(M)$ the family of unions of a circuit family $\mathcal{C}$, any $c \in \mathcal{C}$ would be a nonempty member of $\mathcal{F}(M)$, hence cofinite; it properly contains a cofinite set c' , itself a nonempty member of $\mathcal{F}(M)$ and so a union of a nonempty subfamily of $\mathcal{C}$, which supplies some $c'' \in \mathcal{C}$ with $c'' \subseteq c' \subsetneq c$ — against the antichain axiom. Neither elimination nor (SM) is used. Separately, the sets containing no nonempty support are exactly the coinfinite ones, which have no maximal element, so (SM) fails as well, and there is no minimal nonempty member; (S1) and (S2) hold vacuously. This chain is conditional on Question 9.3; Example 6.2 is not.

7. THE DIAGONAL BLOCK

Definition 7.1. For $M \in \mathcal{P}(\ell^2)$ put

$$\mathcal{D}(M) = \{S \subseteq \mathbb{N} : M \wedge \ell^2(S) \neq 0\}, \quad \mathcal{I}(M) = 2^{\mathbb{N}} \setminus \mathcal{D}(M).$$

Lemma 7.2. $M \wedge \ell^2(S) \neq 0$ if and only if S contains a nonempty member of $\mathcal{F}(M)$. Hence $\mathcal{I}(M)$ consists exactly of the sets containing no nonempty support, and $\mathcal{D}(M)$ is determined by $\mathcal{F}(M)$.

Proof. The meet of closed subspaces is the intersection, and a nonzero $x \in M \cap \ell^2(S)$ is a nonzero $x \in M$ with $\text{supp}(x) \subseteq S$. $\square$

In finite rank, where Theorem 5.1 applies, $\mathcal{I}(M)$ is the family of independent sets of the support matroid: matroid independence is lattice-theoretic disjointness from the block. The minimal members of $\mathcal{D}(M)$ are its circuits, and axiom (SM) asserts that a subspace of the block disjoint from M extends to a maximal such subspace.

Proposition 7.3. For a filter F on $\mathcal{P}(\ell^2)$ with trace $\mathcal{S}(F) = \{S : \ell^2(S) \in F\}$,

$$\mathcal{S}(F) \subseteq \bigcap_{M \in F} \mathcal{D}(M),$$

with equality when F is maximal.

Proof. If $S \in \mathcal{S}(F)$ and $M \in F$ then $M \wedge \ell^2(S) \in F$, which is nonzero by properness, so $S \in \mathcal{D}(M)$. For the converse under maximality, suppose $S \in \mathcal{D}(M)$ for every $M \in F$. Then $\ell^2(S)$ meets every member of F nontrivially, so by the maximality criterion ([FWI], Lemma 2.1) it belongs to F . $\square$

Definition 7.4. M is *diagonally consistent* if some ultrafilter $\mathcal{U}$ on $\mathbb{N}$ satisfies $\mathcal{U} \subseteq \mathcal{D}(M)$. The *covering number* $\chi(M)$ is the least k such that $\mathbb{N}$ is the union of k members of $\mathcal{I}(M)$, and ∞ if there is no such k .

Proposition 7.5. M is diagonally consistent if and only if there is a pluf π with $M \in \pi$ and $\mathcal{S}(\pi)$ an ultrafilter.

Proof. Given $\mathcal{U} \subseteq \mathcal{D}(M)$, let G be the upward closure of the finite meets of $\{M\} \cup \{\ell^2(S) : S \in \mathcal{U}\}$. Such a meet is $M \wedge \ell^2(S_1 \cap \dots \cap S_k)$, nonzero because the intersection lies in $\mathcal{U} \subseteq \mathcal{D}(M)$; so G is proper, and any pluf $\pi \supseteq G$ has $\mathcal{U} \subseteq \mathcal{S}(\pi)$. A trace contains no complementary pair, since $\ell^2(S) \wedge \ell^2(\mathbb{N} \setminus S) = 0$, so $\mathcal{S}(\pi) = \mathcal{U}$. Conversely Proposition 7.3 gives $\mathcal{S}(\pi) \subseteq \mathcal{D}(M)$ for $M \in \pi$. $\square$

Theorem 7.6. M is diagonally consistent if and only if $\chi(M) = \infty$.

Proof. If $\chi(M) = \infty$, the ideal J generated by the downward closed family $\mathcal{I}(M)$ consists of the finite unions of its members and is proper, so its dual filter J^* is proper; extend J^* to an ultrafilter $\mathcal{U}$. For $S \in \mathcal{I}(M)$ we have $\mathbb{N} \setminus S \in J^* \subseteq \mathcal{U}$, so $S \notin \mathcal{U}$; hence $\mathcal{U} \subseteq \mathcal{D}(M)$. Conversely, if $\mathcal{U} \subseteq \mathcal{D}(M)$ and $\mathbb{N} = S_1 \cup \dots \cup S_k$ with every $S_i \in \mathcal{I}(M)$, primeness of $\mathcal{U}$ puts some $S_i \in \mathcal{U} \subseteq \mathcal{D}(M)$, a contradiction. $\square$

Proposition 7.7. M is diagonally consistent if and only if $\mathcal{F}(M) \setminus \{\emptyset\}$ is partition regular: every partition of $\mathbb{N}$ into finitely many cells has a cell containing a nonempty support.

Proof. Theorem 7.6 through Lemma 7.2, covers being refinable to partitions. $\square$

For finite ground sets Edmonds' theorem [E65] computes the covering number as $\max_{A \neq \emptyset} \lceil |A|/r(A) \rceil$; see [Ox11, §11.3].

8. THE COVERING NUMBER IN THE PLUF SERIES

Fix the standard basis. A closed subspace W is *intimate* [FWII] if for every $A \subseteq \mathbb{N}$ at least one of $W \wedge \ell^2(A)$, $W \wedge \ell^2(\mathbb{N} \setminus A)$ is nonzero, and a nonprincipal pluf is *diagonalizable* if its trace is an ultrafilter.

Proposition 8.1. A nonzero closed subspace W is intimate if and only if $\chi(W) \geq 3$.

Proof. W fails to be intimate exactly when some A has both A and $\mathbb{N} \setminus A$ in $\mathcal{I}(W)$, that is, exactly when $\mathbb{N}$ is the union of two members of $\mathcal{I}(W)$. Since $\mathcal{I}(W)$ is downward closed, a cover by two members may be replaced by a complementary pair, so this is $\chi(W) \leq 2$; and $\chi(W) \geq 2$ because $W \neq 0$ puts $\mathbb{N}$ in $\mathcal{D}(W)$. $\square$

Intimacy is thus the first nontrivial value of an invariant whose extreme value is diagonal consistency. The two conditions are different, and Theorem 8.4 separates them; but on plufs they coincide.

Proposition 8.2. Let π be a pluf whose trace is an ultrafilter. Then every member of π is diagonally consistent, hence has $\chi = \infty$.

Proof. Proposition 7.3 gives $\mathcal{S}(\pi) \subseteq \mathcal{D}(M)$ for each $M \in \pi$, and $\mathcal{S}(\pi)$ is an ultrafilter; apply Theorem 7.6. $\square$

Corollary 8.3. *For a nonprincipal pluf the following are equivalent: it is diagonalizable; every member is intimate; every member is diagonally consistent.*

Proof. The first two are equivalent by [FWII], Theorem 5.2. The first implies the third by Proposition 8.2, and the third implies the second by Proposition 8.1. $\square$

The equivalence is a property of filters, not of subspaces. The subspace constructed by Gowers in [FWII] is intimate and has the least covering number compatible with that.

Theorem 8.4. *Let X be the intimate subspace of [FWII], Theorem 4.2. Then $\chi(X) = 3$. Consequently X belongs to no diagonalizable pluf.*

Proof. Write the coordinates in pairs $P_i = \{2i - 1, 2i\}$, $i \geq 1$. By [FWII], Remark 4.4 the membership condition for X is that the sequence of pair sums $s_i = x_{2i-1} + x_{2i}$ be orthogonal to every u_n , and by [FWII], Remark 4.7 the orthogonal complement of $\{u_n\}$ is the line spanned by $w = (1, \frac{1}{2}, \frac{1}{3}, \dots)$. So $x \in X$ if and only if $s = cw$ for some scalar c .

We first identify $\mathcal{I}(X)$. If $S \supseteq P_i$ for some i , then $e_{2i-1} - e_{2i} \in X \wedge \ell^2(S)$, so $S \in \mathcal{D}(X)$. If S meets every P_i in exactly one point a_i , the vector with $x_{a_i} = 1/i$ has pair sums $s_i = 1/i$, hence lies in $X \wedge \ell^2(S)$, so again $S \in \mathcal{D}(X)$. If S contains no P_i and misses some P_j entirely, then any $x \in X \wedge \ell^2(S)$ has $s_j = 0$, forcing $c = 0$ and $s \equiv 0$; then for each i the two coordinates of P_i sum to zero while at most one of them is nonzero, so both vanish, and $x = 0$. Hence

$$\mathcal{I}(X) = \{S : S \text{ contains no } P_i \text{ and misses some } P_j\}.$$

Now $\chi(X) \geq 3$ by Proposition 8.1, X being intimate. For the upper bound, take

$$S = \{1, 6\} \cup \{n \text{ odd} : n \geq 7\}, \quad T = \{2, 3\} \cup \{n \text{ even} : n \geq 8\}, \quad V = \{4, 5\}.$$

These cover $\mathbb{N}$. None contains a complete pair — each meets P_i in at most one point — and each misses one entirely: S misses P_2 , T misses P_3 , and V misses P_1 . So all three lie in $\mathcal{I}(X)$ and $\chi(X) \leq 3$. The last claim is Proposition 8.2. $\square$

Proposition 8.5. *Let $\mathcal{U}$ be a non-selective ultrafilter on $\mathbb{N}$ and let R be an addable blocker in the sense of [FWII], §3: a closed subspace with $\dim(R \wedge \ell^2(S)) = \infty$ for every $S \in \mathcal{U}$. Then $\chi(R) = \infty$.*

Proof. Each $S \in \mathcal{U}$ has $R \wedge \ell^2(S) \neq 0$, so $\mathcal{U} \subseteq \mathcal{D}(R)$; apply Theorem 7.6. $\square$

This is a fourth constraint on the rigidity conjecture of [FWII], independent of the three proved in [FWIV], §6: those bound the dimension of a blocker on finitely many pieces, place its total support in $\mathcal{U}$, and force infinite piece-spread inside every measure-one block, while Proposition 8.5 forbids finite covers of $\mathbb{N}$ by sets disjoint from R .

9. QUESTIONS

Question 9.1. Characterize the support families of closed subspaces of ℓ^2 of infinite rank.

Question 9.2. Does $\mathcal{F}(M)$ satisfy scrawl elimination (S2) when the deleted set is infinite?

Question 9.3. For a Blaschke product Θ , which subsets of $\mathbb{N}_0$ occur as $\{n : \hat{f}(n) \neq 0\}$ for $f \in K_\Theta$? Does K_Θ contain a nonzero function with infinitely many vanishing Taylor coefficients? By Proposition 7.7, K_Θ is diagonally consistent exactly when its family of nonzero coefficient supports is partition regular, so this question decides whether K_Θ can sit in a pluf with ultrafilter trace.

Question 9.4. Which values does $\chi(M)$ take, and how is it computed? Edmonds' formula governs finite ground sets; Theorem 8.4 exhibits the value 3 on an infinite-rank example.

Question 9.5. Which filters on $\mathbb{N}$ arise as traces of plufs? By [FWII], Theorem 3.4 the block filter of $\mathcal{U}$ is itself maximal exactly for selective $\mathcal{U}$; the present question asks for the traces of arbitrary plufs.

Question 9.6. Theorem 5.2 is a statement about $\mathbb{C}$ -representability. Which support families of closed subspaces of the real ℓ^2 arise, and where do the two classifications differ?

Question 9.7. Replace the diagonal masa by another maximal Boolean subalgebra of $\mathcal{P}(\ell^2)$. For the continuous masa the analogous question is the one posed in [FWII], §6.

Formal verification. Sections 2, 3, 7 and 8 have been formalized and machine-checked in Lean 4 with Mathlib (toolchain v4.28.0), extending the development that accompanies [FWI, FWII, FWIII, FWIV]: Lemma 2.1, Proposition 2.2 with its recursion on frozen minima, Corollary 2.3 and Proposition 3.2, whose independence of minimality the mechanization established; Lemma 7.2, both halves of Proposition 7.3, Proposition 7.5 and Theorem 7.6; and all of Section 8 — Propositions 8.1, 8.2 and 8.5, Corollary 8.3, and Theorem 8.4 with the three-cover displayed above. The artifacts compile with no unproved obligations, and every statement’s axiom footprint is exactly `propext`, `Classical.choice`, `Quot.sound`.

The mechanization is carried out over $\mathbb{R}$, which the conventions of Section 1 license: no item required a property of the scalars beyond uncountability of the field, used once, in the counting step behind Lemma 2.1 and Proposition 2.2.

Sections 4–6 are covered as well. Mathlib’s matroid API has no constructor building a matroid from a circuit family, and no scrawls, cofinitary duals or representability; the mechanization therefore defines circuit families, independence, the maximality axiom and scrawl families directly, in terms of families of subsets of $\mathbb{N}$, and imports nothing from that API. Against those definitions: Theorem 4.1 in both clauses; Theorem 5.2, conditional on Theorem 5.1, which is supplied as a named hypothesis about the support families of finite-dimensional subspaces of $\mathbb{R}^{\mathbb{N}}$ and never discharged — the transfer between those and the support families of finite-rank closed subspaces of ℓ^2 , in both directions, is proved, the realization being by a diagonal rescaling with nonvanishing entries; Lemma 6.1, for which a counting form of Rolle’s theorem was built, Mathlib having none; and Example 6.2 and the conditional consequence drawn from it, in the strengthened form recorded in Section 6: the mechanized argument uses only the antichain axiom.

What remains outside the development is what should: Theorem 5.1, which is a theorem of the literature, and Question 9.3, which is open.

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REFERENCES

- [AB15] S. H. Afzali Borujeni and N. Bowler, *Thin sums matroids and duality*, Adv. Math. **271** (2015), 1–29.
- [ABFGGV] F. Aroca, L. Bossinger, S. Falkensteiner, C. Garay López, L. R. González-Ramírez and C. V. Valencia Negrete, *Infinite matroids in tropical differential algebra*, Canad. Math. Bull.; [arXiv:2305.04784](#).
- [BC18] N. Bowler and J. Carmesin, *An excluded minors method for infinite matroids*, J. Combin. Theory Ser. B **128** (2018), 104–113.
- [BDKPW13] H. Bruhn, R. Diestel, M. Kriesell, R. Pendavingh and P. Wollan, *Axioms for infinite matroids*, Adv. Math. **239** (2013), 18–46.
- [E65] J. Edmonds, *Minimum partition of a matroid into independent subsets*, J. Res. Nat. Bur. Standards Sect. B **69B** (1965), 67–72.

- [FWI] D. V. Feldman and A. Wilce, *Ultrafilter limits in the projection lattice of a Hilbert space*, preprint (paper I of this series).
- [FWII] D. V. Feldman and A. Wilce, *Intimate subspaces, block filters, and the diagonalization of maximal projection filters*, preprint (paper II).
- [FWIII] D. V. Feldman and A. Wilce, *Projection-lattice ultrafilters at a measurable cardinal*, preprint (paper III).
- [FWIV] D. V. Feldman and A. Wilce, *Exact paving for Fubini products of normal measures, with an application to maximal projection filters*, preprint (paper IV).
- [GMR16] S. R. Garcia, J. Mashreghi and W. T. Ross, *Introduction to Model Spaces and their Operators*, Cambridge Univ. Press, 2016.
- [K83] G. Kalmbach, *Orthomodular Lattices*, London Math. Soc. Monographs **18**, Academic Press, 1983.
- [MNW14] D. Mayhew, M. Newman and G. Whittle, *Is the missing axiom of matroid theory lost forever?*, Quart. J. Math. **65** (2014), 1397–1415.
- [Ox11] J. Oxley, *Matroid Theory*, 2nd ed., Oxford Univ. Press, 2011.
- [S87] B. Sturmfels, *On the decidability of Diophantine problems in combinatorial geometry*, Bull. Amer. Math. Soc. **17** (1987), 121–124.
- [V78] P. Vámos, *The missing axiom of matroid theory is lost forever*, J. London Math. Soc. **18** (1978), 403–408.

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